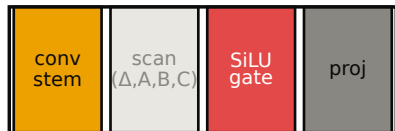


Step 1 -- Freeze the scan (necessity test)

conv stem scan (Δ, A, B, C) SiLU gate proj

BM3-frozen



Step 2 -- Remove gate (localise)

frozen – gate



Step 3 -- Add gate to S4D (sufficiency test)

*S4D + gate
+ added*



$\Delta(\text{BM3-frozen}) \leq 4\text{pp}$ (CI incl. 0)
vs S4D: +15–20pp (full: +21–24)
→ no scan term is primary

gate removal: –9 to –23pp,
to $\approx$ S4D level
→ gate is necessary here

S4D: gate 0.30 $\leq$ additive 0.41
BM3: additive rescues $p \approx 0.01$ –0.02
→ gate-specific in its native block

Conditional on this block, testbed and budget: the gate is necessary as a multiplicative operator in its native block.